INTERNSHIP PROJECT REPORT

(Vulnerability Assessment/Analysis Report)

Organization

ITSOLERA Pvt. Ltd

Department

Offensive Security

Group/Team

Team Alpha

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Vulnerability Assessment Report

Target System:	192.168.56.104
Assessment Type:	Network Vulnerability Assessment
Assessment Tool:	Nmap 7.95
Assessment Date:	April 5th, 2026
Assessor:	Red Team Intern

1. Executive Summary

1.1. Assessment Scope and Objectives

This vulnerability assessment was conducted to evaluate the security posture of the target host at IP address `192.168.56.104`. The scope focused on two open ports: **22 (SSH)** and **3306 (MySQL)**. The primary objective was to identify known vulnerabilities, misconfigurations, and potential entry points that could be exploited by an attacker. The assessment aimed to provide actionable insights for remediation and to strengthen the overall security of the system.

1.2. Testing Narrative

The assessment was performed on *March 12, 2026*, using ***Nmap version 7.95*** with the `-sV` and `--script vuln` options. This command performed service version detection and executed a suite of vulnerability detection scripts against the specified ports. The scan targeted only ports **22** and **3306** to minimize disruption and focus on the services exposed by the host. The results were analyzed to identify vulnerable software versions and associated Common Vulnerabilities and Exposures (CVEs).

1.3. Findings Summary

The scan revealed that both services running on the target host are severely outdated and expose the system to a wide range of critical vulnerabilities. The *OpenSSH* server version `4.7p1` is associated with dozens of vulnerabilities, including multiple remote code execution flaws with CVSS scores as high as ***10.0***, and numerous publicly available exploits. The *MySQL* server version `5.0.51a-3ubuntu5` is also outdated and contains several high-risk vulnerabilities, one with a CVSS score of ***8.5*** and multiple with available exploits. The presence of these vulnerabilities places the host at imminent risk of compromise.

1.4. Remediation Summary

Immediate action is required to mitigate the identified risks. The primary recommendation is to upgrade both ***OpenSSH*** and ***MySQL*** to the latest stable versions that contain security patches. If immediate upgrading is not feasible, temporary workarounds such as restricting access via firewall rules, implementing intrusion detection, and disabling unnecessary features should be applied. A comprehensive patch management policy should be established to prevent similar issues in the future.

2. Assessment Overview

2.1. Assessment Methodology

The assessment employed automated vulnerability scanning using Nmap, a widely used network exploration and security auditing tool. The `vuln` script category was utilized to check for known vulnerabilities in the detected services. This method relies on version

fingerprinting and cross-referencing with vulnerability databases such as CVE, Exploit-DB, and others. The results provide a snapshot of potential weaknesses based on the software versions identified.

2.2. Scan Information

- **Date and Time:** March 12, 2026, 08:29 EDT
- **Tool:** Nmap (Version 7.95)
- **Command:** `nmap -sV --script vuln -p22,3306 192.168.56.104`
- **Duration:** 23.47 seconds

2.3. Target Information

- **IP Address:** 192.168.56.104
- **Host Status:** Up (0.0028s latency)
- **MAC Address:** `08:00:27:7D:43:95` (Oracle VirtualBox virtual NIC)
- **Operating System:** Linux (kernel inferred from service fingerprints)
- **Open Ports and Services:**
  - Port 22/tcp: OpenSSH 4.7p1 Debian 8ubuntu1
  - Port 3306/tcp: MySQL 5.0.51a-3ubuntu5

3. Findings Information

3.1. Scanned Hosts

Only one host was within the scope of this assessment: `192.168.56.104`. The host responded to the scan and both targeted ports were open, indicating active services.

3.2. Types of Vulnerabilities Identified

The vulnerabilities identified can be categorized as follows:

- **Remote Code Execution (RCE):** Multiple flaws in OpenSSH allow an unauthenticated attacker to execute arbitrary code on the target system. These are the most critical, with CVSS scores up to 10.0.
- **Privilege Escalation:** Some vulnerabilities enable a local user to gain elevated privileges, potentially leading to full system compromise.
- **Information Disclosure:** Certain weaknesses could allow an attacker to access sensitive information, such as authentication credentials or system data.

- **Denial of Service (DoS):** Several vulnerabilities can be exploited to crash the service or make it unavailable.
- **Authentication Bypass:** Flaws that allow an attacker to bypass authentication mechanisms and gain unauthorized access.
- **Man-in-the-Middle (MitM):** Some SSH vulnerabilities facilitate MitM attacks, enabling interception and modification of traffic.
- **User Enumeration:** Vulnerabilities that allow an attacker to determine valid usernames on the system, aiding further attacks.

3.3. Detailed Information on Identified Vulnerabilities

3.3.1. OpenSSH 4.7p1

The installed version of OpenSSH is severely outdated and is affected by numerous vulnerabilities. Below are some of the most critical ones identified by the scan:

- **CVE-2023-38408 (CVSS 9.8):** A remote code execution vulnerability in the SSH transport protocol. An unauthenticated attacker can exploit this to execute arbitrary code on the target system.
- **CVE-2016-1908 (CVSS 9.8):** Another RCE flaw that allows an attacker to bypass authentication and execute commands.
- **DF059135-2CF5-5441-8F22-E6EF1DEE5F6E (CVSS 10.0):** An exploit associated with a critical vulnerability that provides a CVSS score of 10.0, indicating maximum severity.
- **CVE-2015-5600 (CVSS 8.5):** A vulnerability that allows remote attackers to cause a denial of service or possibly execute arbitrary code via crafted keyboard-interactive authentication.
- **CVE-2020-15778 (CVSS 7.8):** A command injection flaw in the scp utility that can be exploited by an attacker with valid credentials to execute arbitrary commands.
- **CVE-2018-15473 (CVSS 5.9):** A user enumeration vulnerability that allows an attacker to determine valid usernames by analyzing authentication responses.

The scan output lists dozens of additional CVEs and exploits, many of which have publicly available exploit code. The sheer number of vulnerabilities and the availability of exploits make this service extremely high-risk.

3.3.2. MySQL 5.0.51a-3ubuntu5

The MySQL version is also outdated and contains several security flaws:

- **SSV:19118 (CVSS 8.5):** A vulnerability that could allow an attacker to gain unauthorized access or cause a denial of service. This entry from Seebug includes

exploit code.

- **CVE-2017-15945 (CVSS 7.8):** A privilege escalation vulnerability that allows a local user to gain elevated privileges.
- **CVE-2009-4028 (CVSS 6.8):** A flaw in the handling of certain requests that could lead to arbitrary code execution or denial of service.
- **CVE-2010-1621 (CVSS 5.0):** A buffer overflow vulnerability that could be exploited to cause a denial of service or potentially execute code.
- **CVE-2008-2079 (CVSS 4.6):** A privilege escalation vulnerability that allows authenticated users to gain higher privileges.

These vulnerabilities, combined with the age of the software, indicate that the MySQL service is susceptible to various attacks, including unauthorized data access, denial of service, and potential code execution.

3.4. Notes Describing Additional Details of Scan Results

- The scan identified a large number of vulnerabilities for both services, many of which are marked with `*EXPLOIT*`, indicating that working exploit code is publicly available. This significantly increases the likelihood of a successful attack.
- The target appears to be a virtual machine (based on the MAC address OUI), possibly a test or legacy system. However, if this host is part of a production network, it poses a substantial risk.
- No additional services were scanned; the assessment was limited to ports ***22*** and ***3306***. Other ports may also host vulnerable services, but they were outside the scope.

4. Risk Assessment

4.1. Classification of Vulnerabilities Based on Risk Level

Identified vulnerabilities are classified according to their CVSS scores & potential impact:

- **Critical (CVSS 9.0-10.0):** Multiple OpenSSH vulnerabilities, including those with CVSS 10.0 and 9.8, fall into this category. These flaws allow remote code execution without authentication and have public exploits.
- **High (CVSS 7.0-8.9):** Several OpenSSH vulnerabilities (e.g., CVE-2015-5600, CVSS 8.5) and the MySQL SSV:19118 (CVSS 8.5) are classified as high risk. They can lead to privilege escalation, denial of service, or code execution under certain conditions.
- **Medium (CVSS 4.0-6.9):** This group includes information disclosure flaws (e.g., CVE-2018-15473) and some MySQL vulnerabilities (e.g., CVE-2009-4028, CVSS 6.8).
- **Low (CVSS 0-3.9):** Some minor vulnerabilities, such as CVE-2008-3259 (CVSS 1.2) for

SSH and CVE-2010-1626 (CVSS 3.6) for MySQL, pose limited risk but should still be addressed as part of a comprehensive hardening effort.

4.2. Potential Vulnerabilities That Can Compromise the System

The most significant threats to the target system are the remote code execution vulnerabilities in OpenSSH. An attacker exploiting these could gain complete control over the host without any authentication. Once inside, they could pivot to other systems on the network, steal data, install malware, or disrupt operations. The MySQL vulnerabilities, while generally requiring some level of access, could lead to data breaches or privilege escalation, further aiding an attacker in maintaining persistence or exfiltrating sensitive information.

4.3. Critical Hosts with Severe Vulnerabilities

The host `192.168.56.104` is considered critical due to the presence of multiple high-severity vulnerabilities in both exposed services. The combination of outdated software and publicly available exploits makes this host an attractive target for attackers. If this host is accessible from untrusted networks (such as the internet), the risk of compromise is extremely high. Even if it is only on an internal network, an attacker who gains a foothold elsewhere could use these vulnerabilities to move laterally.

5. Recommendations

5.1. Prioritization of Remediation Based on the Risk Ranking

Remediation efforts should be prioritized as follows:

- **Immediate (within 24-48 hours):** Address critical vulnerabilities in OpenSSH, particularly those with CVSS scores of 9.0 and above. If patching is not immediately possible, implement compensating controls such as network-level access restrictions.
- **Short-term (within one week):** Address high-risk vulnerabilities in both OpenSSH and MySQL, including those with available exploits.
- **Medium-term (within one month):** Address medium-risk vulnerabilities and conduct a full security review of the host.
- **Long-term (ongoing):** Establish processes for regular patching and vulnerability assessments.

5.2. Action Plan to Implement the Recommendations for Each Identified Vulnerability

- **OpenSSH Upgrade:** Upgrade OpenSSH to the latest version (or at least a version that addresses the identified CVEs). For Debian-based systems, this typically involves running `apt-get update && apt-get upgrade openssh-server`. Verify the new version and test connectivity before deploying to production.

- **MySQL Upgrade:** Upgrade MySQL to a supported version. Given the age of the current version, it may be necessary to migrate to a newer release (e.g., MySQL 5.7 or 8.0) or consider using a fork like MariaDB. Ensure compatibility with any applications relying on the database.
- **If Upgrading Is Not Immediately Possible:**
  - Restrict access to SSH & MySQL via firewall rules (e.g., allow only specific IPs).
  - Disable unused features (e.g., SSH port forwarding, MySQL remote access) to reduce attack surface.
  - Implement intrusion detection systems (IDS) to monitor for exploitation attempts.
  - Apply vendor-provided workarounds or patches if available.

5.3. Root-Cause Analysis

The root cause of the vulnerabilities is the use of outdated software versions. ***OpenSSH 4.7p1*** was released in 2007, and ***MySQL 5.0.51a*** dates back to 2008. These versions are no longer supported by their vendors and contain numerous unpatched security flaws. The lack of a regular patch management process has allowed these vulnerabilities to persist, exposing the system to significant risk.

5.4. Application of Patches/Fixes

- For OpenSSH, obtain the latest package from the official distribution repositories or compile from source if necessary. Ensure that all dependencies are also updated.
- For MySQL, back up all databases, then upgrade to a supported version following the official upgrade guides. Test the upgrade in a non-production environment first.
- After patching, re-run vulnerability scans to confirm that the issues have been resolved.

5.5. Lessons Learned

This assessment highlights the critical importance of maintaining up-to-date software. Legacy systems often become security liabilities because they lack patches for known vulnerabilities. Organizations should enforce a policy that mandates regular updates and decommissioning of unsupported software. Additionally, periodic vulnerability assessments can catch such issues before they are exploited.

5.6. Awareness Training

System administrators and IT staff should receive training on the importance of patch management and secure configuration. They should be made aware of the risks associated with running outdated software and the steps to mitigate those risks. Training should also cover how to use vulnerability scanning tools and interpret their results.

5.7. Implementation of Periodic Vulnerability Assessment

A schedule for regular vulnerability assessments should be established. For critical systems, scans should be performed at least monthly, or whenever significant changes are made. Automated tools can help streamline this process. The results should be reviewed and acted upon promptly.

5.8. Implementation of Policies, Procedures, and Controls

- **Patch Management Policy:** Define a formal policy that requires timely application of security patches. Include timelines based on risk severity (e.g., critical patches within 48 hours).
- **Change Management:** Ensure that all changes, including software upgrades, follow a change management process to minimize disruption.
- **Access Controls:** Implement the principle of least privilege for both SSH and MySQL access. Use firewalls and VPNs to restrict remote access.
- **Configuration Hardening:** Develop and enforce security baseline configurations for all services. For SSH, disable root login, use key-based authentication, and limit allowed users. For MySQL, remove anonymous accounts, disable remote root access, and use strong passwords.
- **Incident Response:** Update incident response plans to include procedures for handling exploitation of known vulnerabilities.

By implementing these recommendations, the organization can significantly reduce the risk posed by the identified vulnerabilities and improve its overall security posture.

6. Conclusion

The vulnerability assessment of host 192.168.56.104 has revealed a critically dangerous security posture due to the use of severely outdated software versions for both OpenSSH and MySQL. The presence of numerous high-severity vulnerabilities, many with publicly available exploit code, places this system at imminent risk of complete compromise. Immediate remediation actions, starting with the upgrading of both services, are essential to mitigate these risks. Beyond patching, this assessment underscores the necessity of establishing a robust patch management policy, conducting regular vulnerability scans, and fostering a security-aware culture among IT staff. Without swift intervention, the host could serve as an entry point for attackers to infiltrate the broader network, leading to data breaches, service disruptions, and reputational damage. The findings serve as a stark reminder that security maintenance is not optional but a critical component of operational integrity.

7. Appendices

7.1. Appendix A: Raw Nmap Scan Output

The following is the complete output from the Nmap vulnerability scan conducted on March 12, 2026. It includes all detected vulnerabilities and exploit references for both OpenSSH and MySQL services.

```
┌──(kali㉿kali)-[~]
└─$ nmap -sV --script vuln -p22,3306 192.168.56.104
Starting Nmap 7.95 ( https://nmap.org ) at 2026-03-12 08:29 EDT
Nmap scan report for 192.168.56.104
Host is up (0.0028s latency).

PORT     STATE SERVICE VERSION
22/tcp   open  ssh     OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
| vulners:
|   cpe:/a:openbsd:openssh:4.7p1:
|       DF059135-2CF5-5441-8F22-E6EF1DEE5F6E    10.0
https://vulners.com/gitee/DF059135-2CF5-5441-8F22-E6EF1DEE5F6E  *EXPLOIT*
|       PACKETSTORM:173661      9.8
https://vulners.com/packetstorm/PACKETSTORM:173661      *EXPLOIT*
|       F0979183-AE88-53B4-86CF-3AF0523F3807    9.8
https://vulners.com/githubexploit/F0979183-AE88-53B4-86CF-3AF0523F3807  *EXPLOIT*
|       CVE-2023-38408  9.8     https://vulners.com/cve/CVE-2023-38408
|       CVE-2016-1908   9.8     https://vulners.com/cve/CVE-2016-1908
|       B8190CDB-3EB9-5631-9828-8064A1575B23    9.8
https://vulners.com/githubexploit/B8190CDB-3EB9-5631-9828-8064A1575B23  *EXPLOIT*
|       8FC9C5AB-3968-5F3C-825E-E8DB5379A623    9.8
https://vulners.com/githubexploit/8FC9C5AB-3968-5F3C-825E-E8DB5379A623  *EXPLOIT*
|       8AD01159-548E-546E-AA87-2DE89F3927EC    9.8
https://vulners.com/githubexploit/8AD01159-548E-546E-AA87-2DE89F3927EC  *EXPLOIT*
|       6192C35D-F78B-5C0A-AB8D-9826A79A5320    9.8
https://vulners.com/githubexploit/6192C35D-F78B-5C0A-AB8D-9826A79A5320  *EXPLOIT*
|       2227729D-6700-5C8F-8930-1EEAFD4B9FF0    9.8
https://vulners.com/githubexploit/2227729D-6700-5C8F-8930-1EEAFD4B9FF0  *EXPLOIT*
|       0221525F-07F5-5790-912D-F4B9E2D1B587    9.8
https://vulners.com/githubexploit/0221525F-07F5-5790-912D-F4B9E2D1B587  *EXPLOIT*
|       CVE-2015-5600   8.5     https://vulners.com/cve/CVE-2015-5600
|       BA3887BD-F579-53B1-A4A4-FF49E953E1C0    8.1
https://vulners.com/githubexploit/BA3887BD-F579-53B1-A4A4-FF49E953E1C0  *EXPLOIT*
|       4FB01B00-F993-5CAF-BD57-D7E290D10C1F    8.1
https://vulners.com/githubexploit/4FB01B00-F993-5CAF-BD57-D7E290D10C1F  *EXPLOIT*
|       SSV:78173       7.8     https://vulners.com/seebug/SSV:78173    *EXPLOIT*
|       SSV:69983       7.8     https://vulners.com/seebug/SSV:69983    *EXPLOIT*
|       PACKETSTORM:98796       7.8     https://vulners.com/packetstorm/PACKETSTORM:98796
*EXPLOIT*
|       PACKETSTORM:94556       7.8     https://vulners.com/packetstorm/PACKETSTORM:94556
*EXPLOIT*
|       PACKETSTORM:101052      7.8
https://vulners.com/packetstorm/PACKETSTORM:101052      *EXPLOIT*
|       EXPLOITPACK:71D51B69AA2D3A74753D7A921EE79985    7.8
https://vulners.com/exploitpack/EXPLOITPACK:71D51B69AA2D3A74753D7A921EE79985    *EXPLOIT*
|       EXPLOITPACK:67F6569F63A082199721C069C852BBD7    7.8
https://vulners.com/exploitpack/EXPLOITPACK:67F6569F63A082199721C069C852BBD7    *EXPLOIT*
|       EDB-ID:24450    7.8     https://vulners.com/exploitdb/EDB-ID:24450      *EXPLOIT*
|       EDB-ID:15215    7.8     https://vulners.com/exploitdb/EDB-ID:15215      *EXPLOIT*
|       CVE-2020-15778  7.8     https://vulners.com/cve/CVE-2020-15778
|       CVE-2016-10012  7.8     https://vulners.com/cve/CVE-2016-10012
|       C94132FD-1FA5-5342-B6EE-0DAF45EEFFE3    7.8
https://vulners.com/githubexploit/C94132FD-1FA5-5342-B6EE-0DAF45EEFFE3  *EXPLOIT*
|       991D2CC4-0E09-5745-97A2-4917461BD6EC    7.8
```

```
https://vulners.com/githubexploit/991D2CC4-0E09-5745-97A2-4917461BD6EC  *EXPLOIT*
|       2E719186-2FED-58A8-A150-762EFBAAA523    7.8
https://vulners.com/gitee/2E719186-2FED-58A8-A150-762EFBAAA523  *EXPLOIT*
|       23CC97BE-7C95-513B-9E73-298C48D74432    7.8
https://vulners.com/githubexploit/23CC97BE-7C95-513B-9E73-298C48D74432  *EXPLOIT*
|       10213DBE-F683-58BB-B6D3-353173626207    7.8
https://vulners.com/githubexploit/10213DBE-F683-58BB-B6D3-353173626207  *EXPLOIT*
|       SSV:92579       7.5     https://vulners.com/seebug/SSV:92579    *EXPLOIT*
|       SSV:61450       7.5     https://vulners.com/seebug/SSV:61450    *EXPLOIT*
|       CVE-2016-10708  7.5     https://vulners.com/cve/CVE-2016-10708
|       CVE-2016-10009  7.5     https://vulners.com/cve/CVE-2016-10009
|       CVE-2014-1692   7.5     https://vulners.com/cve/CVE-2014-1692
|       CVE-2010-4478   7.5     https://vulners.com/cve/CVE-2010-4478
|       CF52FA19-B5DB-5D14-B50F-2411851976E2    7.5
https://vulners.com/githubexploit/CF52FA19-B5DB-5D14-B50F-2411851976E2  *EXPLOIT*
|       1337DAY-ID-26576        7.5     https://vulners.com/zdt/1337DAY-ID-26576
*EXPLOIT*
|       SSV:92582       7.2     https://vulners.com/seebug/SSV:92582    *EXPLOIT*
|       CVE-2016-10010  7.0     https://vulners.com/cve/CVE-2016-10010
|       SSV:92580       6.9     https://vulners.com/seebug/SSV:92580    *EXPLOIT*
|       CVE-2015-6564   6.9     https://vulners.com/cve/CVE-2015-6564
|       1337DAY-ID-26577        6.9     https://vulners.com/zdt/1337DAY-ID-26577
*EXPLOIT*
|       EDB-ID:46516    6.8     https://vulners.com/exploitdb/EDB-ID:46516      *EXPLOIT*
|       EDB-ID:46193    6.8     https://vulners.com/exploitdb/EDB-ID:46193      *EXPLOIT*
|       CVE-2019-6110   6.8     https://vulners.com/cve/CVE-2019-6110
|       CVE-2019-6109   6.8     https://vulners.com/cve/CVE-2019-6109
|       1337DAY-ID-32328        6.8     https://vulners.com/zdt/1337DAY-ID-32328
*EXPLOIT*
|       1337DAY-ID-32009        6.8     https://vulners.com/zdt/1337DAY-ID-32009
*EXPLOIT*
|       D104D2BF-ED22-588B-A9B2-3CCC562FE8C0    6.5
https://vulners.com/githubexploit/D104D2BF-ED22-588B-A9B2-3CCC562FE8C0  *EXPLOIT*
|       CVE-2023-51385  6.5     https://vulners.com/cve/CVE-2023-51385
|       CVE-2008-1657   6.5     https://vulners.com/cve/CVE-2008-1657
|       C07ADB46-24B8-57B7-B375-9C761F4750A2    6.5
https://vulners.com/githubexploit/C07ADB46-24B8-57B7-B375-9C761F4750A2  *EXPLOIT*
|       A88CDD3E-67CC-51CC-97FB-AB0CACB6B08C    6.5
https://vulners.com/githubexploit/A88CDD3E-67CC-51CC-97FB-AB0CACB6B08C  *EXPLOIT*
|       65B15AA1-2A8D-53C1-9499-69EBA3619F1C    6.5
https://vulners.com/githubexploit/65B15AA1-2A8D-53C1-9499-69EBA3619F1C  *EXPLOIT*
|       5325A9D6-132B-590C-BDEF-0CB105252732    6.5
https://vulners.com/gitee/5325A9D6-132B-590C-BDEF-0CB105252732  *EXPLOIT*
|       530326CF-6AB3-5643-AA16-73DC8CB44742    6.5
https://vulners.com/githubexploit/530326CF-6AB3-5643-AA16-73DC8CB44742  *EXPLOIT*
|       EDB-ID:40858    6.4     https://vulners.com/exploitdb/EDB-ID:40858      *EXPLOIT*
|       EDB-ID:40119    6.4     https://vulners.com/exploitdb/EDB-ID:40119      *EXPLOIT*
|       EDB-ID:39569    6.4     https://vulners.com/exploitdb/EDB-ID:39569      *EXPLOIT*
|       CVE-2016-3115   6.4     https://vulners.com/cve/CVE-2016-3115
|       PACKETSTORM:181223      5.9
https://vulners.com/packetstorm/PACKETSTORM:181223      *EXPLOIT*
|       MSF:AUXILIARY-SCANNER-SSH-SSH_ENUMUSERS-        5.9
https://vulners.com/metasploit/MSF:AUXILIARY-SCANNER-SSH-SSH_ENUMUSERS- *EXPLOIT*
|       EDB-ID:45939    5.9     https://vulners.com/exploitdb/EDB-ID:45939      *EXPLOIT*
|       EDB-ID:45233    5.9     https://vulners.com/exploitdb/EDB-ID:45233      *EXPLOIT*
|       CVE-2023-48795  5.9     https://vulners.com/cve/CVE-2023-48795
|       CVE-2019-6111   5.9     https://vulners.com/cve/CVE-2019-6111
|       CVE-2018-15473  5.9     https://vulners.com/cve/CVE-2018-15473
|       721F040C-37BC-59E1-9433-01A2EAC2E755    5.9
```

```
https://vulners.com/githubexploit/721F040C-37BC-59E1-9433-01A2EAC2E755  *EXPLOIT*
|       SSV:61911       5.8     https://vulners.com/seebug/SSV:61911    *EXPLOIT*
|       EXPLOITPACK:98FE96309F9524B8C84C508837551A19    5.8
https://vulners.com/exploitpack/EXPLOITPACK:98FE96309F9524B8C84C508837551A19    *EXPLOIT*
|       EXPLOITPACK:5330EA02EBDE345BFC9D6DDDD97F9E97    5.8
https://vulners.com/exploitpack/EXPLOITPACK:5330EA02EBDE345BFC9D6DDDD97F9E97    *EXPLOIT*
|       CVE-2014-2653   5.8     https://vulners.com/cve/CVE-2014-2653
|       CVE-2014-2532   5.8     https://vulners.com/cve/CVE-2014-2532
|       SSV:91041       5.5     https://vulners.com/seebug/SSV:91041    *EXPLOIT*
|       PACKETSTORM:140019      5.5
https://vulners.com/packetstorm/PACKETSTORM:140019      *EXPLOIT*
|       PACKETSTORM:136251      5.5
https://vulners.com/packetstorm/PACKETSTORM:136251      *EXPLOIT*
|       PACKETSTORM:136234      5.5
https://vulners.com/packetstorm/PACKETSTORM:136234      *EXPLOIT*
|       EXPLOITPACK:F92411A645D85F05BDBD274FD222226F    5.5
https://vulners.com/exploitpack/EXPLOITPACK:F92411A645D85F05BDBD274FD222226F    *EXPLOIT*
|       EXPLOITPACK:9F2E746846C3C623A27A441281EAD138    5.5
https://vulners.com/exploitpack/EXPLOITPACK:9F2E746846C3C623A27A441281EAD138    *EXPLOIT*
|       EXPLOITPACK:1902C998CBF9154396911926B4C3B330    5.5
https://vulners.com/exploitpack/EXPLOITPACK:1902C998CBF9154396911926B4C3B330    *EXPLOIT*
|       CVE-2016-10011  5.5     https://vulners.com/cve/CVE-2016-10011
|       1337DAY-ID-25388        5.5     https://vulners.com/zdt/1337DAY-ID-25388
*EXPLOIT*
|       FD18B68B-C0A6-562E-A8C8-781B225F15B0    5.3
https://vulners.com/githubexploit/FD18B68B-C0A6-562E-A8C8-781B225F15B0  *EXPLOIT*
|       E9EC0911-E2E1-52A7-B2F4-D0065C6A3057    5.3
https://vulners.com/githubexploit/E9EC0911-E2E1-52A7-B2F4-D0065C6A3057  *EXPLOIT*
|       CVE-2018-20685  5.3     https://vulners.com/cve/CVE-2018-20685
|       CVE-2017-15906  5.3     https://vulners.com/cve/CVE-2017-15906
|       CVE-2016-20012  5.3     https://vulners.com/cve/CVE-2016-20012
|       CNVD-2018-20962 5.3     https://vulners.com/cnvd/CNVD-2018-20962
|       CNVD-2018-20960 5.3     https://vulners.com/cnvd/CNVD-2018-20960
|       A9E6F50E-E7FC-51D0-9C93-A43461469FA2    5.3
https://vulners.com/githubexploit/A9E6F50E-E7FC-51D0-9C93-A43461469FA2  *EXPLOIT*
|       A801235B-9835-5BA8-B8FE-23B7FFCABD66    5.3
https://vulners.com/githubexploit/A801235B-9835-5BA8-B8FE-23B7FFCABD66  *EXPLOIT*
|       8DD1D813-FD5A-5B26-867A-CE7CAC9FEEDF    5.3
https://vulners.com/gitee/8DD1D813-FD5A-5B26-867A-CE7CAC9FEEDF  *EXPLOIT*
|       4F2FBB06-E601-5EAD-9679-3395D24057DD    5.3
https://vulners.com/githubexploit/4F2FBB06-E601-5EAD-9679-3395D24057DD  *EXPLOIT*
|       486BB6BC-9C26-597F-B865-D0E904FDA984    5.3
https://vulners.com/githubexploit/486BB6BC-9C26-597F-B865-D0E904FDA984  *EXPLOIT*
|       2385176A-820F-5469-AB09-C340264F2B2F    5.3
https://vulners.com/gitee/2385176A-820F-5469-AB09-C340264F2B2F  *EXPLOIT*
|       1337DAY-ID-31730        5.3     https://vulners.com/zdt/1337DAY-ID-31730
*EXPLOIT*
|       SSV:60656       5.0     https://vulners.com/seebug/SSV:60656    *EXPLOIT*
|       SSH_ENUM        5.0     https://vulners.com/canvas/SSH_ENUM     *EXPLOIT*
|       PACKETSTORM:150621      5.0
https://vulners.com/packetstorm/PACKETSTORM:150621      *EXPLOIT*
|       EXPLOITPACK:F957D7E8A0CC1E23C3C649B764E13FB0    5.0
https://vulners.com/exploitpack/EXPLOITPACK:F957D7E8A0CC1E23C3C649B764E13FB0    *EXPLOIT*
|       EXPLOITPACK:EBDBC5685E3276D648B4D14B75563283    5.0
https://vulners.com/exploitpack/EXPLOITPACK:EBDBC5685E3276D648B4D14B75563283    *EXPLOIT*
|       CVE-2010-5107   5.0     https://vulners.com/cve/CVE-2010-5107
|       CVE-2015-5352   4.3     https://vulners.com/cve/CVE-2015-5352
|       CVE-2010-4755   4.0     https://vulners.com/cve/CVE-2010-4755
|       CVE-2021-36368  3.7     https://vulners.com/cve/CVE-2021-36368
```

```
|       CVE-2025-61985  3.6     https://vulners.com/cve/CVE-2025-61985
|       CVE-2025-61984  3.6     https://vulners.com/cve/CVE-2025-61984
|       B7EACB4F-A5CF-5C5A-809F-E03CCE2AB150    3.6
https://vulners.com/githubexploit/B7EACB4F-A5CF-5C5A-809F-E03CCE2AB150  *EXPLOIT*
|       4C6E2182-0E99-5626-83F6-1646DD648C57    3.6
https://vulners.com/githubexploit/4C6E2182-0E99-5626-83F6-1646DD648C57  *EXPLOIT*
|       CVE-2012-0814   3.5     https://vulners.com/cve/CVE-2012-0814
|       CVE-2011-5000   3.5     https://vulners.com/cve/CVE-2011-5000
|       SSV:92581       2.1     https://vulners.com/seebug/SSV:92581    *EXPLOIT*
|       CVE-2011-4327   2.1     https://vulners.com/cve/CVE-2011-4327
|       CVE-2015-6563   1.9     https://vulners.com/cve/CVE-2015-6563
|       CVE-2008-3259   1.2     https://vulners.com/cve/CVE-2008-3259
|       PACKETSTORM:151227      0.0
https://vulners.com/packetstorm/PACKETSTORM:151227      *EXPLOIT*
|       PACKETSTORM:140261      0.0
https://vulners.com/packetstorm/PACKETSTORM:140261      *EXPLOIT*
|       1337DAY-ID-30937        0.0     https://vulners.com/zdt/1337DAY-ID-30937
*EXPLOIT*
|       1337DAY-ID-26468        0.0     https://vulners.com/zdt/1337DAY-ID-26468
*EXPLOIT*
|       1337DAY-ID-25391        0.0     https://vulners.com/zdt/1337DAY-ID-25391
*EXPLOIT*
|       1337DAY-ID-20301        0.0     https://vulners.com/zdt/1337DAY-ID-20301
*EXPLOIT*
|_      1337DAY-ID-14373        0.0     https://vulners.com/zdt/1337DAY-ID-14373
*EXPLOIT*
3306/tcp open  mysql   MySQL 5.0.51a-3ubuntu5
|_ssl-ccs-injection: No reply from server (TIMEOUT)
| vulners:
|   cpe:/a:mysql:mysql:5.0.51a-3ubuntu5:
|       SSV:19118       8.5     https://vulners.com/seebug/SSV:19118    *EXPLOIT*
|       CVE-2017-15945  7.8     https://vulners.com/cve/CVE-2017-15945
|       SSV:15006       6.8     https://vulners.com/seebug/SSV:15006    *EXPLOIT*
|       CVE-2009-4028   6.8     https://vulners.com/cve/CVE-2009-4028
|       SSV:15004       6.0     https://vulners.com/seebug/SSV:15004    *EXPLOIT*
|       CVE-2010-1621   5.0     https://vulners.com/cve/CVE-2010-1621
|       CVE-2015-2575   4.9     https://vulners.com/cve/CVE-2015-2575
|       SSV:3280        4.6     https://vulners.com/seebug/SSV:3280     *EXPLOIT*
|       CVE-2008-2079   4.6     https://vulners.com/cve/CVE-2008-2079
|       CVE-2010-3682   4.0     https://vulners.com/cve/CVE-2010-3682
|       CVE-2010-3677   4.0     https://vulners.com/cve/CVE-2010-3677
|       CVE-2009-0819   4.0     https://vulners.com/cve/CVE-2009-0819
|       CVE-2007-5925   4.0     https://vulners.com/cve/CVE-2007-5925
|_      CVE-2010-1626   3.6     https://vulners.com/cve/CVE-2010-1626
MAC Address: 08:00:27:7D:43:95 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at
https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 23.47 seconds

┌──(kali㉿kali)-[~]
└─$
```

7.2. Appendix B: Nmap Scan Screenshots

```
Session Actions Edit View Help
┌──(kali㉿kali)-[~]
└─$ nmap -sV --script vuln -p22,3306 192.168.56.104
Starting Nmap 7.95 ( https://nmap.org ) at 2026-03-12 08:29 EDT
Nmap scan report for 192.168.56.104
Host is up (0.0028s latency).
```

Figure 1: Generic Information about the command Used to run and Scan the target system

```
PORT     STATE SERVICE VERSION
22/tcp   open  ssh     OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
| vulners:
|   cpe:/a:openbsd:openssh:4.7p1:
|       DF059135-2CF5-5441-8F22-E6EF1DEE5F6E    10.0    https://vulners.com/gitee/DF059135-2CF5-5441-8F22-E6EF1DEE5F6E  *EXPLOIT*
|       PACKETSTORM:173661      9.8     https://vulners.com/packetstorm/PACKETSTORM:173661      *EXPLOIT*
|       F0979183-AE88-53B4-86CF-3AF0523F3807    9.8     https://vulners.com/githubexploit/F0979183-AE88-53B4-86CF-3AF0523F3807  *EXPLOIT*
|       CVE-2023-38408  9.8     https://vulners.com/cve/CVE-2023-38408
|       CVE-2016-1908   9.8     https://vulners.com/cve/CVE-2016-1908
|       B8190CDB-3EB9-5631-9828-8064A1575B23    9.8     https://vulners.com/githubexploit/B8190CDB-3EB9-5631-9828-8064A1575B23  *EXPLOIT*
|       8FC9C5AB-3968-5F3C-825E-E8DB5379A623    9.8     https://vulners.com/githubexploit/8FC9C5AB-3968-5F3C-825E-E8DB5379A623  *EXPLOIT*
|       8AD01159-548E-546E-AA87-2DE89F3927EC    9.8     https://vulners.com/githubexploit/8AD01159-548E-546E-AA87-2DE89F3927EC  *EXPLOIT*
|       6192C35D-F78B-5C0A-AB8D-9826A79A5320    9.8     https://vulners.com/githubexploit/6192C35D-F78B-5C0A-AB8D-9826A79A5320  *EXPLOIT*
|       2227729D-6700-5C8F-8930-1EEAFD4B9FF0    9.8     https://vulners.com/githubexploit/2227729D-6700-5C8F-8930-1EEAFD4B9FF0  *EXPLOIT*
|       0221525F-07F5-5790-912D-F4B9E2D1B587    9.8     https://vulners.com/githubexploit/0221525F-07F5-5790-912D-F4B9E2D1B587  *EXPLOIT*
|       CVE-2015-5600   8.5     https://vulners.com/cve/CVE-2015-5600
|       BA3887BD-F579-53B1-A4A4-FF49E953E1C0    8.1     https://vulners.com/githubexploit/BA3887BD-F579-53B1-A4A4-FF49E953E1C0  *EXPLOIT*
|       4FB01B00-F993-5CAF-BD57-D7E290D10C1F    8.1     https://vulners.com/githubexploit/4FB01B00-F993-5CAF-BD57-D7E290D10C1F  *EXPLOIT*
|       SSV:78173       7.8     https://vulners.com/seebug/SSV:78173    *EXPLOIT*
|       SSV:69983       7.8     https://vulners.com/seebug/SSV:69983    *EXPLOIT*
|       PACKETSTORM:98796       7.8     https://vulners.com/packetstorm/PACKETSTORM:98796       *EXPLOIT*
|       PACKETSTORM:94556       7.8     https://vulners.com/packetstorm/PACKETSTORM:94556       *EXPLOIT*
|       PACKETSTORM:101052      7.8     https://vulners.com/packetstorm/PACKETSTORM:101052      *EXPLOIT*
|       EXPLOITPACK:71D51B69AA2D3A74753D7A921EE79985    7.8     https://vulners.com/exploitpack/EXPLOITPACK:71D51B69AA2D3A74753D7A921EE799
|       EXPLOITPACK:67F6569F63A082199721C069C852BBD7    7.8     https://vulners.com/exploitpack/EXPLOITPACK:67F6569F63A082199721C069C852BE
|       EDB-ID:24450    7.8     https://vulners.com/exploitdb/EDB-ID:24450      *EXPLOIT*
|       EDB-ID:15215    7.8     https://vulners.com/exploitdb/EDB-ID:15215      *EXPLOIT*
|       CVE-2020-15778  7.8     https://vulners.com/cve/CVE-2020-15778
|       CVE-2016-10012  7.8     https://vulners.com/cve/CVE-2016-10012
|       C94132FD-1FA5-5342-B6EE-0DAF45EEFFE3    7.8     https://vulners.com/githubexploit/C94132FD-1FA5-5342-B6EE-0DAF45EEFFE3  *EXPLOIT*
```

Figure 2: Information about the Exploits (on Exploit-DB) and CVE-IDs against SSH 4.7p1.

```
3306/tcp open  mysql   MySQL 5.0.51a-3ubuntu5
|_ssl-ccs-injection: No reply from server (TIMEOUT)
| vulners:
|   cpe:/a:mysql:mysql:5.0.51a-3ubuntu5:
|       SSV:19118        8.5     https://vulners.com/seebug/SSV:19118     *EXPLOIT*
|       CVE-2017-15945   7.8     https://vulners.com/cve/CVE-2017-15945
|       SSV:15006        6.8     https://vulners.com/seebug/SSV:15006     *EXPLOIT*
|       CVE-2009-4028    6.8     https://vulners.com/cve/CVE-2009-4028
|       SSV:15004        6.0     https://vulners.com/seebug/SSV:15004     *EXPLOIT*
|       CVE-2010-1621    5.0     https://vulners.com/cve/CVE-2010-1621
|       CVE-2015-2575    4.9     https://vulners.com/cve/CVE-2015-2575
|       SSV:3280         4.6     https://vulners.com/seebug/SSV:3280      *EXPLOIT*
|       CVE-2008-2079    4.6     https://vulners.com/cve/CVE-2008-2079
|       CVE-2010-3682    4.0     https://vulners.com/cve/CVE-2010-3682
|       CVE-2010-3677    4.0     https://vulners.com/cve/CVE-2010-3677
|       CVE-2009-0819    4.0     https://vulners.com/cve/CVE-2009-0819
|       CVE-2007-5925    4.0     https://vulners.com/cve/CVE-2007-5925
|_      CVE-2010-1626    3.6     https://vulners.com/cve/CVE-2010-1626
MAC Address: 08:00:27:7D:43:95 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel
```

Figure 3: Information about the CVE-IDs found against the MySQL 5.0.51a-3ubuntu5.

THE END